
Distilling solvent response into reusable priors for simulation-free hydration prediction

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Abstract

Physical simulations expose how a solvent responds to a molecule, but usually repeat that calculation for every query. We ask whether heterogeneous solvation responses can instead be learned once from structure and reused as molecular coordinates. SolvAI distills 15 calculated, empirical, corrected, and conformational responses into structure-predicted priors, then combines them with standard molecular descriptors in an experimentally supervised hydration-free-energy endpoint. Deployment requires only a SMILES string: no molecular dynamics, path-integral dynamics, or probe calculation is run. In a preregistered matched comparison on the 85 neutral ARROW solutes, adding the aligned priors lowers mean absolute error from 0.303 to 0.202 kcal mol⁻¹; five complete partitions give 0.207 ± 0.004. Shuffling molecule-prior alignment abolishes the gain. The advantage survives chemical separation, removal of all ARROW endpoint labels, and a prospectively frozen 220-molecule external cohort, where error falls from 1.532 to 1.153 kcal mol⁻¹. A negative experiment with sparse PIMD responses shows that physical fidelity alone is insufficient: transfer requires a response that is relevant, structure-predictable, and non-redundant. Simulation outputs can thus become reusable supervision rather than a per-query computation.

1 Can solvent simulation have a life beyond one query?

Hydration free energy is the reversible work of transferring a molecule from gas into water. It links molecular structure to partitioning, binding, and reactivity, while testing electrostatics, polarization, dispersion, repulsion, and solvent reorganization. Explicit alchemical simulations estimate it by sampling solvent response along a coupling path [19]; path-integral molecular dynamics (PIMD) can additionally represent nuclear delocalization [10]. The ARROW study makes the resulting accuracy-cost tradeoff concrete: across 85 neutral solutes, its reconstructed classical MAE is 0.785 kcal mol⁻¹ and eight-bead PIMD reaches 0.205 kcal mol⁻¹ [13]. Both require a new simulation for each molecule.

Learned hydration models occupy several positions along this spectrum. Direct graph or fingerprint models learn the experimental endpoint; multi-fidelity and Δ -learning approaches transfer between calculated and measured targets [11, 15, 16, 22]. Other hybrids retain a physical calculation at inference, for example an alchemical, continuum, or integral-equation descriptor [1, 2, 18, 21]. A complementary strategy predicts quantum or physical descriptors before using them in a downstream task [6, 20]. These precedents motivate a sharper question: can complementary descriptions of *solvent response* be consolidated into a structure-predictable layer whose information survives after the underlying calculation disappears?

SolvAI tests this question, not whether one network can imitate a particular trajectory engine. Six external source families define 15 compact coordinates: COSMOtherm water response; five Abraham

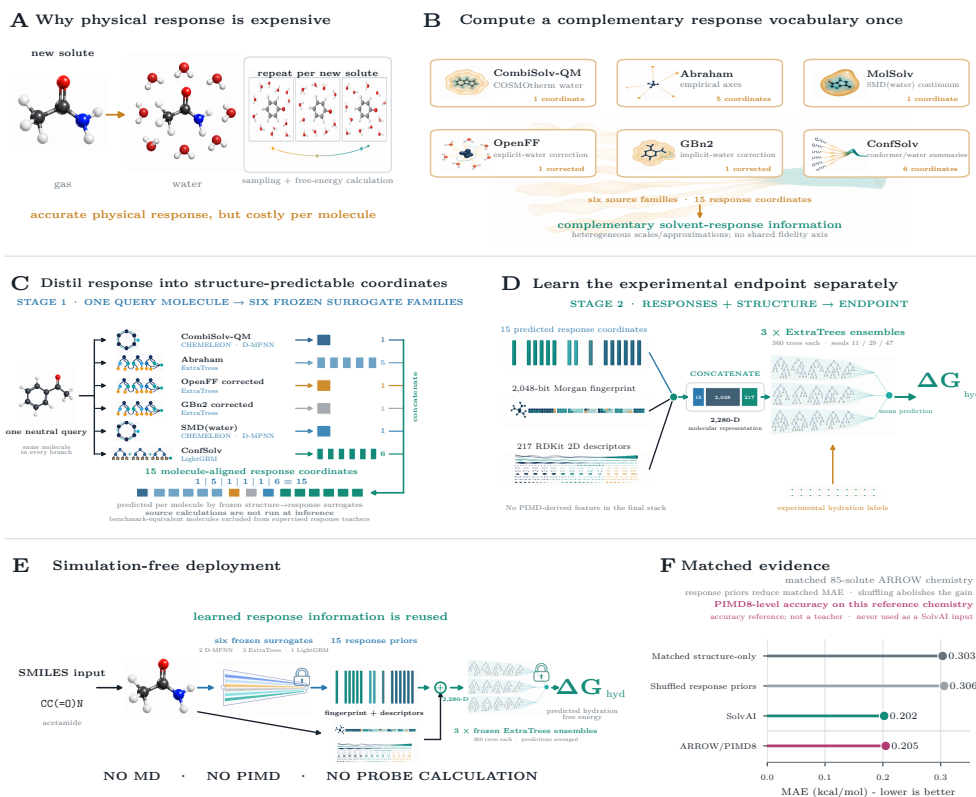


Figure 1: **Solvent response becomes reusable molecular supervision.** External calculations and measurements define 15 response coordinates. Structure-to-response surrogates predict them in stage 1; a matched experimental endpoint combines them with ordinary descriptors in stage 2. Deployment repeats only learned mappings. ARROW/PIMDS is an accuracy reference, never a retained teacher.

response axes; residual-corrected explicit- and implicit-solvent predictions; SMD(water); and six summaries of conformational reweighting and water-response distributions [3, 7–9, 12, 14, 22]. Independent stage-1 surrogates learn each coordinate from molecular structure. A stage-2 endpoint combines their predictions with a 2,048-bit Morgan fingerprint and 217 RDKit descriptors (Fig. 1). The deployment input is only a SMILES string. No MD, PIMD, continuum, alchemical, or probe calculation runs for a new query. The scientific test is therefore whether predicted response adds information beyond a capable structure-only endpoint, not whether SolvAI can access expensive calculations at inference.

2 What exactly is distilled?

The response layer is deliberately heterogeneous. Its coordinates are not asserted to be additive components of hydration free energy, nor are their source fidelities put on one artificial scale. Instead they provide complementary views of solute–water interaction: bulk continuum response, hydrogen-bond acidity and basicity, polarizability, corrected explicit- and implicit-solvent estimates, and conformational changes between gas and water. Stage 2 is allowed to learn which combinations are useful under experimental supervision.

The external sources also differ greatly in statistical support. The MolSolv archive contains 1,729,545 SMD calculations; filtering, deduplication, and standardized identity exclusion leave 350,359 structures for the confirmatory teacher. ConfSolv contains 5,392,567 conformer records for 43,693 solutes; target availability, quality filtering, and identity controls leave 17,829 structures for its response model. Smaller sources contribute 3,959 CombiSolv-QM structures, 8,098 Abraham records, 520

corrected OpenFF records, and 550 corrected GBn2 records. These counts describe source-specific training support, not a pooled dataset.

Every coordinate is an explicit prediction made before endpoint fitting. The final 2,280-dimensional endpoint input contains 15 response priors, 2,048 fingerprint bits, and 217 RDKit descriptors. This explicit interface makes the scientific hypothesis destructively testable: the response block can be removed, permuted while preserving its internal covariance, or substituted by alternative outputs without changing the endpoint learner. It also prevents the stage-1 models from silently consuming an experimental hydration value at query time.

Identity control is applied to the teachers as well as the endpoint pool. An initial exact-connectivity filter is followed by cleanup, largest-fragment selection, uncharging, and canonical-tautomer enumeration. The expanded audit discovers 2 additional CombiSolv-QM, 32 MolSolv, and 22 ConfSolv equivalents; they are removed and the affected teachers are refitted without changing the remaining source splits. The reported analysis is the corrected run only.

3 Do aligned response priors add information beyond structure?

We isolate the response layer by comparing two endpoints with identical experimental labels, molecular descriptors, architecture, weights, folds, and random seeds. The only difference is the 15 predicted priors. The 1,280-molecule external experimental pool contains no exact or standardized-equivalent benchmark identity. Exact, fragment-parent, uncharged-parent, and canonical-tautomer audits also remove benchmark equivalents from every supervised response source before the affected teachers are refitted. Thus stage 1 cannot copy the held-out benchmark identity through a source table.

On the preregistered five-fold ARROW-85 evaluation, the matched structure endpoint has MAE 0.303 kcal mol⁻¹; the complete response stack reaches 0.202 kcal mol⁻¹ and improves 53 of 85 solutes (Fig. 2). Its error is on the same scale as ARROW/PIMD8 (0.205 kcal mol⁻¹), used solely as a high-fidelity simulation reference. This is not a claim that the two methods are equivalent: one is experimentally supervised and the paired comparison does not establish superiority over PIMD8.

Two destructive controls identify the source of the gain. First, permuting all 15 columns as one block preserves their joint distribution while breaking molecule alignment; five permutations return MAE to 0.306, near the structure-only control. Second, five complete fold assignments give mean MAE 0.313 for structure only and 0.207 ± 0.004 for SolvAI, with the aligned model better in every partition. Useful information therefore resides in the molecule-specific predicted response, not arbitrary dimensionality or one fortunate split.

4 Where is the reusable signal carried?

The block progression separates a compact core from complementary refinements. The empirical and residual-corrected response block is independently positive. Adding the compact stack—CombiSolv-QM, five Abraham axes, and corrected OpenFF/GBn2 responses—moves the endpoint from the structure-only baseline to 0.212 kcal mol⁻¹; adding SMD(water) gives 0.207, and the six conformational-response summaries complete the 0.202 result. Computation-derived, SMD, and ConfSolv blocks are weaker when tested alone, but their joint addition is positive. The pattern is evidence for complementarity among predicted coordinates, not an energy decomposition and not a claim that every prior contributes equally to every molecule.

This distinction matters because the endpoint already receives a high-capacity two-dimensional molecular representation. A response prior is valuable only if it makes a solvent-conditioned relation easier to recover than it is from fingerprint and RDKit coordinates alone. The improvement of 0.101 kcal mol⁻¹ has a paired 95% interval from -0.215 to -0.020 and survives the one-factor ARROW-weight sensitivity. Conversely, replacing aligned priors by block-permuted values erases the improvement even though the endpoint sees an equally sized and equally correlated feature block. The evidence concerns information alignment, not feature count.

The stage-1 outputs should also not be read as hidden access to benchmark simulation. The retained teachers are trained on benchmark-disjoint source records after the expanded identity audit. ARROW classical, PIMD4, and PIMD8 values never enter the final response vector or endpoint fit. They remain external reference predictions on the same 85 molecules. The comparison is consequently

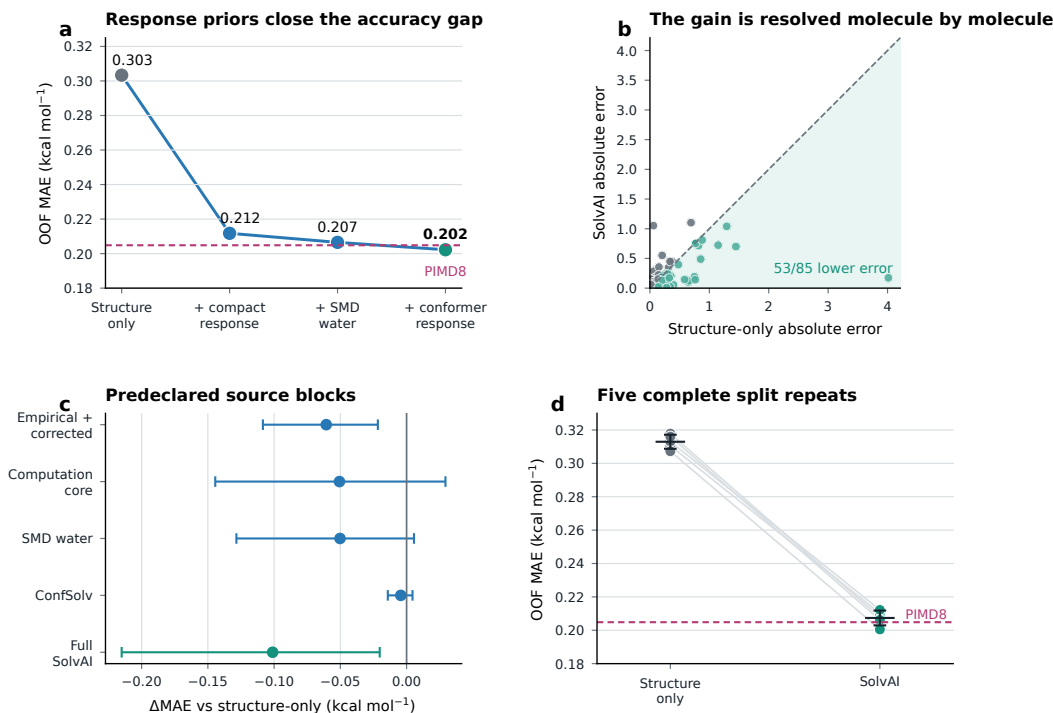


Figure 2: **Molecule-aligned response priors improve a matched endpoint.** **a**, Frozen response-block progression and the separate PIMD8 reference. **b**, Molecule-level paired absolute errors. **c**, Predeclared block effects with 95% molecule-bootstrap intervals. **d**, Five complete split repeats. Every endpoint contrast changes only the response block.

between two distinct routes: an endpoint that reuses learned response coordinates with experimental supervision, and a high-fidelity trajectory method whose value here is to set a demanding error scale.

Finally, prediction of an intermediate creates an auditable scientific interface. Each of the 15 coordinates has a source definition, unit, training population, surrogate, and frozen artifact hash. This is less flexible than an unrestricted end-to-end latent vector, but it makes removal, shuffling, source-disjoint auditing, and failed-teacher analysis possible. The experiments below exploit that interface to ask whether the response signal persists as endpoint supervision becomes more remote.

5 Does the signal survive separation from nearby labels?

Random folds can leave related molecules in the endpoint pool. We therefore apply each separation rule to *all* endpoint-supervised rows. For every outer test partition, public and ARROW training molecules in the held-out cluster, scaffold, or functional family are removed. A complementary analysis removes any training row at or above a specified Morgan similarity to a test molecule.

The response contrast survives every predeclared regime (Fig. 3). At maximum allowed train-test similarity 0.70, MAE falls from 0.343 to 0.211 kcal mol⁻¹; cluster separation lowers it from 0.349 to 0.216. Scaffold and functional-family separation are substantially harder, but still favour SolvAI: 0.728 to 0.376 and 1.262 to 0.468. Every paired 95% interval is below zero. Absolute errors in the hardest regimes make clear that the priors improve extrapolation without solving it.

Removing all 85 ARROW experimental labels from endpoint training gives a second transport test. Trained only on the same 1,280 external labels, structure-only error is 0.385 and SolvAI error is 0.257 kcal mol⁻¹. Because the method was developed in the context of ARROW chemistry, this is not labelled an independent external benchmark.

The independent test is a prospectively frozen cohort derived from original measurements in a Henry-law compilation [17]. Eligibility is fixed without model errors. The 220 neutral solutes have

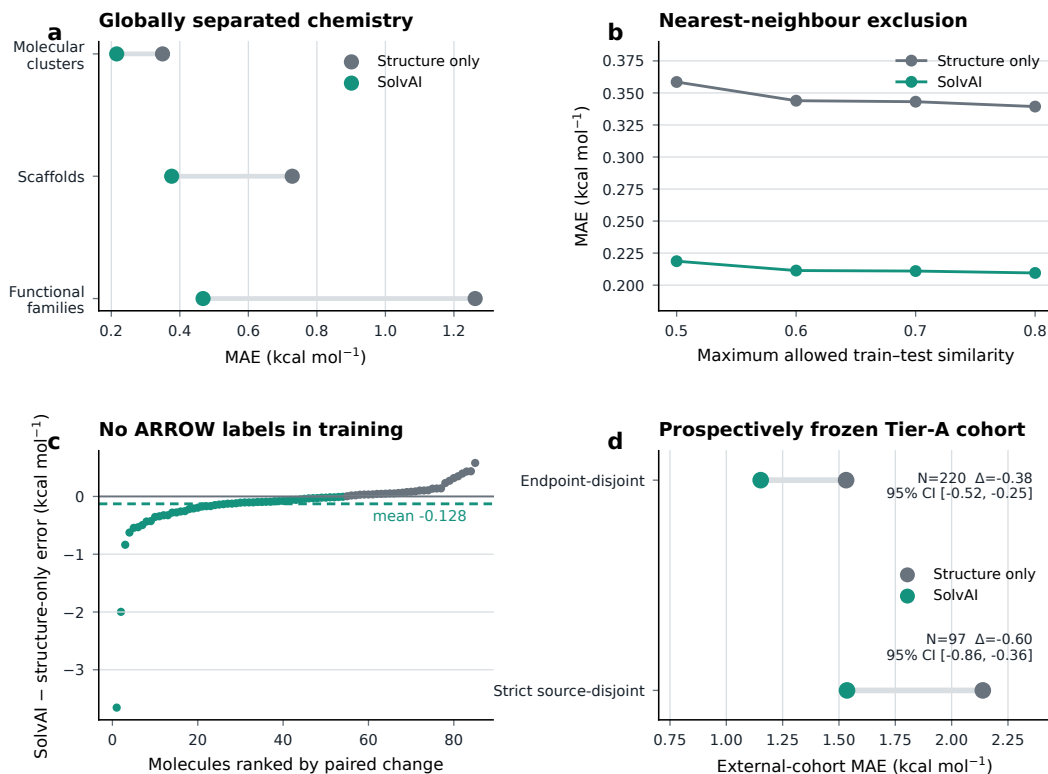


Figure 3: **The response-prior advantage survives chemical separation.** **a**, Cluster, scaffold, and functional-family separation. **b**, Maximum train-test Morgan-similarity exclusions. **c**, molecule-level change without ARROW labels. **d**, The frozen external cohort and its response-source-disjoint subset. Intervals summarize paired absolute-error differences; negative favours SolvAI.

Table 1: **Matched response-layer contrasts across evaluation regimes.** Values are MAE in kcal mol^{-1} . Repeated partitions report the mean across five complete partitions. All other rows use their frozen evaluation.

Evaluation	<i>N</i>	Structure only	SolvAI
ARROW fixed partition	85	0.303	0.202
Five complete partitions	85	0.313	0.207
Maximum similarity 0.70	85	0.343	0.211
Cluster separation	85	0.349	0.216
Scaffold separation	85	0.728	0.376
Functional-family separation	85	1.262	0.468
Zero-ARROW training	85	0.385	0.257
Frozen external cohort	220	1.532	1.153
Response-source-disjoint	97	2.138	1.536

no identity equivalent in endpoint training; a strict 97-molecule subset is additionally absent from all six supervised response sources. On the complete cohort, matched MAE falls from 1.532 to 1.153 kcal mol^{-1} ; on the strict subset it falls from 2.138 to 1.536. The larger absolute errors delimit applicability: the external result transfers the response-layer advantage, not the PIMD8-scale accuracy observed on ARROW-85.

6 What do the controls rule out?

Each evaluation attacks a different shortcut. The matched ablation holds the endpoint learner, labels, ordinary descriptors, weights, folds, and seeds fixed, ruling out an architectural or data-volume

139 explanation for the primary contrast. Block shuffling then preserves response-column marginals and
140 correlations but destroys which molecule owns each vector; its return to structure-only accuracy rules
141 out generic regularization from appending 15 columns. Repeated partitions show that neither result
142 depends on the released fold assignment.

143 Chemical separation addresses interpolation through nearby endpoint labels. Because exclusions
144 are applied to all 1,280 public labels and the available ARROW outer-training rows, a held-out
145 scaffold or family cannot remain in a secondary endpoint source. Zero-ARROW training removes
146 fold-local adaptation to the reference set altogether. These tests do not make ARROW-85 an
147 independent benchmark, but they show that the response contrast survives progressively weaker
148 endpoint proximity.

149 The frozen external cohort tests endpoint transport. Its measurement source, molecular identities,
150 qualification rules, and exclusions were fixed before SolvAI errors were generated. The complete
151 cohort removes identity equivalents from all endpoint labels; the strict subset also removes any
152 equivalent from each response-teacher source. The remaining gain cannot be explained by the same
153 molecule appearing under another string in either learning stage. Its larger errors prevent a stronger
154 but unsupported claim of uniform external accuracy.

155 These controls form a ladder rather than redundant replications. Together they test feature necessity,
156 molecule alignment, split stability, distance from endpoint labels, and source-level identity separation.
157 What remains is the intended mechanism: a structure-predicted solvent-response coordinate can carry
158 endpoint-relevant information that the standard molecular representation does not expose as readily.

159 7 Which physical responses are transferable?

160 Adding physically motivated outputs is not sufficient. Compact ConfSolv summaries transfer better
161 than graph or feed-forward latent representations trained on the same source (Fig. 4a); several force-
162 field, pair-interaction, and implicit-solvent candidates are neutral or harmful under matched screens.
163 These tests do not rank the underlying simulators. They ask whether each output can be inferred from
164 deployed structure accurately enough to add non-redundant endpoint information.

165 Sparse two-bead PIMD calculations at $\lambda = 0.1, 0.5$, and 0.9 expose this bottleneck. Structure-
166 only heads recover broad trends but have component errors of $1.27\text{--}5.20\text{ kcal mol}^{-1}$. Adding the
167 predicted PIMD2 response worsens the matched campaign endpoint from 0.196 to $0.201\text{ kcal mol}^{-1}$;
168 a predicted classical-NQE-PIMD hierarchy reaches 0.215 , their combination 0.219 , and direct
169 integration of the predicted curve yields 1.51 . No PIMD-derived coordinate is retained in SolvAI.

170 Together the positive and negative experiments suggest a three-part condition. A physical response
171 must be relevant to the endpoint, predictable from the representation available at deployment, and
172 non-redundant with other inputs. Higher-fidelity physics can be less useful than a coarser coordinate
173 if sparse data make its response unrecoverable. The transferable object is not “simulation” in the
174 abstract, but an aligned response that survives structure prediction.

175 8 What is gained operationally?

176 The computational saving is structural rather than a claim about a particular GPU kernel. A physical
177 calculation is paid for while constructing a teacher source, then amortized across later queries through
178 the learned response map. On the release host, the complete CPU pipeline has a warm median of
179 15.29 s for one molecule and 15.82 s for a batch of 32 (0.494 s per molecule). This is not a millisecond
180 predictor, but it removes trajectory generation, equilibration, coupling-window sampling, and PIMD
181 replicas that otherwise recur for each solute. Comparisons with a published 15-window PIMD8
182 workload therefore concern total workflow, not same-hardware kernel speed.

183 The explicit response bottleneck offers a second operational benefit: it identifies where new compu-
184 tation is valuable. Adding more experimental endpoint labels tests only the last mapping. Adding
185 source calculations improves a response teacher and can benefit every endpoint that consumes that
186 coordinate. Conversely, the failed PIMD2 experiment says that sparse high-fidelity labels are currently
187 a poor investment for this interface. A targeted set of roughly 250–500 chemically diverse molecules
188 with complete λ -resolved total, electrostatic, polarization, and dispersion/repulsion response would
189 test whether that source can cross the structure-predictability threshold.

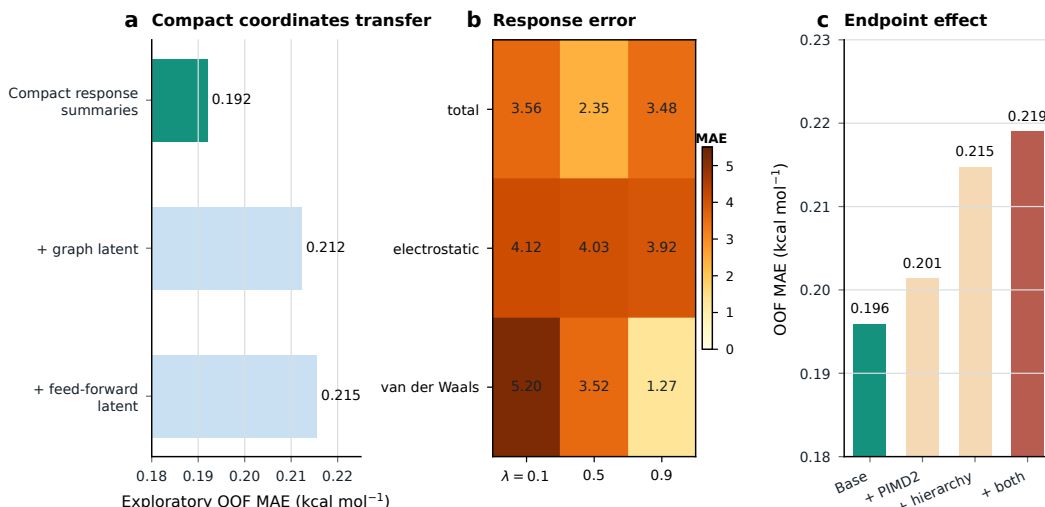


Figure 4: **Transfer is limited by structure-to-response accuracy.** **a**, Compact ConfSolv summaries versus latent coordinates from the same source. **b**, OOF errors of predicted PIMD2 response components. **c**, Their endpoint consequences. The negative result motivates the transfer condition; none of these PIMD-derived blocks enters the final model.

9 Discussion and limitations

SolvAI changes the unit of reuse for physical computation. Rather than spending a simulation once to predict one molecule, it treats responses collected across many molecules as supervision for a reusable coordinate system. Matched ablation, shuffled priors, global chemical separation, zero-ARROW transport, and frozen external evaluation all point to the same conclusion: predicted solvent response contributes information beyond ordinary molecular features and endpoint labels.

The claim is narrower than universal hydration prediction. The primary set has 85 neutral small molecules, and five partitions support an accuracy *scale*, not a guaranteed sub-0.20 threshold. Accuracy degrades under scaffold and functional-family separation and on the external cohort. The current model is unvalidated for ions, salts, biomacromolecules, and substantially larger chemistry; it has no calibrated per-query applicability score. Its experimentally supervised endpoint should not be described as a replacement for PIMD where mechanistic trajectories, nuclear quantum observables, or accuracy outside this domain are required.

The negative PIMD2 result is also a constructive boundary. Direct alchemical-response distillation will require a larger protocol-matched set spanning response curves and interaction components, not merely more endpoint labels. More generally, a simulator is a promising teacher when it exposes a compact response that is predictable across chemistry and complementary to structure. That criterion offers a way to choose which expensive calculations are worth amortizing into scientific models.

10 Methods in brief

The primary reference comprises the 85 neutral ARROW solutes at 298 K. The endpoint pool contains 1,147 connectivity-resolved FreeSolv/CombiSolv-Exp records and 133 multi-source SoluteML records after benchmark exclusion [3, 4, 24]. Fifteen response coordinates are predicted by six frozen artifact families: CHEMELEON-initialized D-MPNNs for CombiSolv-QM and MolSolv [23], ExtraTrees models for Abraham and corrected OpenFF/GBn2 targets [5], and LightGBM models for six ConfSolv summaries. The endpoint averages three 360-tree ExtraTrees pipelines. The matched structure-only model removes only the response columns.

The main protocol uses a frozen five-fold partition, followed by five complete partitions. Block shuffling preserves covariance among the 15 priors. Chemical separation is applied to the complete supervised pool using deterministic functional families, Bemis–Murcko scaffolds, Taylor–Butina clusters, or maximum Morgan similarity. MAE in kcal mol⁻¹ is primary; all comparisons are paired

220 by molecule. Percentile intervals use 100,000 bootstrap resamples with a predeclared seed. The
221 external cohort is never used for teacher, feature, threshold, or endpoint selection. Appendix Secs. S1–
222 S15 provide full source processing, model settings, identity audit, fold assignments, sensitivity
223 analyses, tables, runtime, hashes, and molecule-level results.

224 Available ARROW outer-training labels receive weight 3 and external endpoint rows weight 1 in the
225 frozen protocol. A one-factor sensitivity changing only this weight to 1 leaves the contrast essentially
226 unchanged: 0.310 for structure only and 0.206 kcal mol⁻¹ for SolvAI, with paired change -0.103 and
227 a 95% interval from -0.219 to -0.020. The all-data deployment artifact is refitted only after evaluation
228 is frozen and is never used to report paper accuracy. Its SMILES canonicalization does not imply
229 invariance to tautomer, protonation, or salt form.

230 The external labels are derived from original-measurement Henry-law records at 298.15 K using
231 $K_c = H_s^{cp} RT$ and $\Delta G_{\text{hyd}} = -RT \ln K_c$, with 1 M ideal-gas and 1 M ideal-dilute aqueous
232 standard states. Qualification requires a connected neutral solute, supported elements, compatible
233 target metadata, and no standardized identity equivalent in the endpoint pool. One independently
234 documented name–registry–structure conflict is excluded before evaluation, leaving the 220 reported
235 molecules. No external label participates in fitting, feature choice, thresholding, or model selection.

236 **Data and code availability**

237 An anonymized artifact containing benchmark records, held-out predictions, source manifests, split
238 assignments, audit tables, frozen inference artifacts, and reproduction scripts will accompany the sub-
239 mission. Third-party datasets are referenced by immutable identifiers and hashes where redistribution
240 is restricted. The public repository address is omitted during double-blind review.

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Appendix

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Supplementary Methods

S1. Reference-set processing

The ARROW reference table was reconstructed from machine-readable project outputs and contains 85 unique neutral solutes in water at 298 K. One molecule-condition row was retained for each connectivity. The reference includes experimental hydration free energy and, where available in the source, uncertainty, classical ARROW, PIMD4 and PIMD8 values. Molecular names were not used for identity. Canonical isomeric SMILES, full InChIKey and the first InChIKey block were generated with RDKit 2026.03.5. The released benchmark table is `data/benchmark/arrow_solvation_master.parquet`.

The reconstructed ARROW/PIMD8 MAE is 0.204836 kcal mol⁻¹ and the classical ARROW MAE is 0.784648 kcal mol⁻¹. These values are direct errors against the same experimental column used for SolvAI evaluation; neither simulation value is an input to the final model.

S2. External endpoint labels

The endpoint-label catalog initially contained 5,075 benchmark-disjoint connectivities. A frozen source-quality rule retained all 1,147 rows from an earlier FreeSolv/CombiSolv-Exp merge and 133 additional SoluteML dGsolvDB3 rows supported by at least two source measurements, giving 1,280 external labels. Duplicate structures were resolved at the connectivity level before this rule was applied. The source composition is listed in Supplementary Table 3. The raw merged tables are not redistributed where source terms are not explicit; URLs, versions, hashes and reconstruction commands are given in `repro/DATA_PROVENANCE.md`.

For a random outer fold, these 1,280 rows were combined with the available ARROW outer-training molecules. External rows received sample weight 1 and ARROW outer-training rows weight 3. The experimental value of an outer-test molecule was never used. In the zero-ARROW transfer experiment, only the 1,280 external labels were fitted and all 85 ARROW molecules were evaluated.

The weight of 3 was part of the prospective confirmatory protocol rather than a post-result choice. A separately committed one-factor sensitivity changed only the available ARROW outer-training weight to 1. Under this setting, matched structure-only and full SolvAI MAEs are 0.30977 and 0.20642 kcal mol⁻¹, respectively; the paired change is -0.10335 kcal mol⁻¹ (95% interval, -0.21883 to -0.01995).

S3. Response-source processing

CombiSolv-QM. The water subset was deduplicated by solute connectivity and expressed in kcal mol⁻¹. After the original exact-connectivity exclusion, 3,961 structures were available. Conservative canonical-tautomer comparison found two additional benchmark equivalents; the confirmatory teacher therefore used 3,959 rows.

SoluteML Abraham parameters. Five Abraham solute axes (E, S, A, B and L) were predicted from 8,098 benchmark-disjoint structures. These coordinates respectively describe excess molar refraction, dipolarity/polarizability, hydrogen-bond acidity, hydrogen-bond basicity and hexadecane-air partition response. No standardized benchmark equivalent was found in the audited source.

OpenFF explicit-water response. The source contains 520 benchmark-disjoint absolute solvation-free-energy records from an OpenFF 2.3.0/NAGL/TIP3P protocol with 14 alchemical states. Separate ExtraTrees models predicted calculated hydration free energy and its experimental residual. Their sum is the retained corrected OpenFF coordinate.

GBn2 implicit response. The source contains 550 benchmark-disjoint GBn2 hydration calculations. Separate ExtraTrees models predicted the calculated value and its experimental residual. The sum is the retained corrected implicit-solvent coordinate.

MolSolv SMD(water). The source archive contains 1,729,545 M06-2X/6-31G* SMD calculations. The frozen selection retained all eligible neutral structures with at most 10 heavy atoms and a deterministic one-in-eight sample for structures with 11–18 heavy atoms. After deduplication and exact-connectivity filtering, 350,391 rows remained. Canonical tautomer standardization identified 32 additional benchmark-equivalent records. The confirmatory model used 350,359 rows while keeping every retained molecule in its original source split.

ConfSolv H₂O. The archive contains 5,392,567 H₂O conformer records for 43,693 solutes. We converted kJ mol⁻¹ to kcal mol⁻¹ and aggregated conformers into gas, solution and hydration conformer corrections together with water-response distribution summaries. The broader quality filter produced 39,878 neutral connectivities. The selected teacher target set had 17,851 rows with complete features and admissible energy range; fragment-parent, uncharging and tautomer standardization removed 22 additional benchmark-equivalent records, leaving 17,829.

S4. The 15 response priors

Supplementary Table 1 defines every retained coordinate. At inference, the table’s “surrogate” column is the calculation that runs; the expensive source method does not. The final vector contains one CombiSolv-QM response, five Abraham axes, one corrected OpenFF response, one corrected GBn2 response, one SMD(water) response and six ConfSolv summaries, for 15 coordinates in total.

The arithmetic is important. Richer CombiSolv, OpenFF, GBn2 and ConfSolv targets were available and were evaluated during model development. Only the listed coordinates belong to the released stack. In particular, no PIMD2, PIMD4, PIMD8, $dH/d\lambda$ or NQE coordinate is present.

S5. Surrogate models

The CombiSolv-QM and MolSolv targets were fitted with Chemprop 2.2.4 directed-message-passing models initialized from CHEMELEON. Both used a two-layer feed-forward head of width 256, dropout 0.1, initial/maximum/final learning rates of 3×10^{-5} , 10^{-4} and 10^{-5} , and PyTorch seed 20260826. CombiSolv-QM used RDKit 2D features, batch size 64, a 30-epoch cap and patience 7; MolSolv used the molecular graph, batch size 256, a 25-epoch cap and patience 6. The original frozen train/validation/test memberships were 80/10/10% for CombiSolv-QM and 90/5/5% for MolSolv. Standardized equivalents were removed from their existing split rather than resplitting the remaining source.

Abraham, OpenFF and GBn2 targets used ExtraTrees regressors on the same 2,265 structure features used by the endpoint. The selected leaf sizes and target-level OOF errors are recorded in the source metadata. ConfSolv used 217 RDKit descriptors and the 256 highest-variance Morgan bits. Each target was fitted with a LightGBM L1-regression model with 450 estimators, learning rate 0.04, 31 leaves, minimum child size 20, row subsampling 0.8, column subsampling 0.55 and L_2 regularization 0.1. Target-specific seeds were 20260826 plus the target index for validation and 1000 plus that index for the all-retained-source fit.

S6. Endpoint model

The molecular representation comprises a 2,048-bit radius-2 Morgan fingerprint and 217 RDKit 2D descriptors. The confirmatory endpoint appends the 15 response priors, giving 2,280 columns. A median imputer is fitted on each training partition and adds missingness indicators only where required. Each of three **ExtraTreesRegressor** models uses 360 trees, squared-error splitting, `max_features=0.7`, `min_samples_leaf=2`, `min_samples_split=2`, unlimited depth and no bootstrap; seeds are 11, 29 and 47. Predictions are averaged. The matched no-prior model removes only the 15 response columns.

After all evaluation was frozen, the release artifact was refitted on the 1,280 external labels and all 85 ARROW labels using the same weights and configuration. That refit supplies deployment predictions only. It is not used to calculate any reported held-out score.

S7. Validation partitions

The fixed primary result uses the released `fold_random` assignment. Five complete repeated five-fold partitions use shuffled K-fold seeds 314159, 271828, 161803, 141421 and 173205. All comparisons use the same fold, training rows and model seeds within a partition.

For functional-family separation, SMARTS rules were applied in the precedence order carboxylic acid, amide, ester, aldehyde, ketone, alcohol/phenol, ether, amine, thiol, sulfide/disulfide, aromatic heterocycle, carbocyclic aromatic, alkene/diene, saturated hydrocarbon and other. A test-fold family was removed from both public and ARROW endpoint training rows. Bemis–Murcko scaffold separation used canonical isomeric scaffold SMILES; acyclic molecules were keyed by primary family. Molecular clusters were constructed once on the combined endpoint pool with Taylor–Butina clustering at Morgan Tanimoto similarity 0.70. Nearest-neighbour analyses excluded training rows at similarity thresholds 0.50, 0.60, 0.70 and 0.80 to any outer-test molecule.

S8. Shuffled-prior control

All 15 response columns were permuted together to preserve their joint distribution. For each outer fold, public, ARROW outer-training and ARROW outer-test response rows were permuted separately. The experimental target was never used to define a permutation. Five base seeds (88001–88005) were used for the fixed partition and each repeated partition. This control changes molecule–response alignment while retaining feature count, response marginal distributions and inter-prior correlations.

S9. Statistical analysis

MAE is primary. RMSE, median absolute error, R^2 and the fraction of molecules improved are secondary. Paired candidate-minus-reference differences were resampled at molecule level 100,000 times with seed 20260828; percentile 2.5th and 97.5th quantiles define reported intervals. For the five repeated partitions, each molecule’s absolute error was first averaged over partitions before paired resampling. The preregistration defined a positive comparison as a complete interval below zero and a material effect as an absolute MAE change of at least 0.010 kcal mol^{−1}. No correction was applied to convert the predeclared panel into one binary discovery claim; all comparisons are reported.

S10. Tier-A external validation

The original 221-molecule Tier-A cohort was frozen in a separate ASP-19/GB validation project before its model predictions. It was derived from type-M original-measurement records in Sander’s Henry-law compilation v5.0.0. Intrinsic pure-water H_s^{cp} values normalized to 298.15 K were converted as $K_c = H_s^{cp} RT$ and $\Delta G_{\text{hyd}} = -RT \ln K_c$. This produces the same 1 M gas-to-1 M aqueous standard-state convention used for the endpoint. The source values are printed to two significant figures.

SolvAI qualification was performed without reading SolvAI predictions or errors. All 221 rows passed the connected, neutral, supported-element and target-compatibility rules. One candidate was excluded because the earlier ASP audit had documented a definite conflict between its registry name/CAS and supplied InChIKey structure. The remaining 220 molecules had no exact, fragment-parent, uncharged-parent or canonical-tautomer equivalent among the 1,280 external or 85 ARROW endpoint labels. Of these, 123 occurred in at least one supervised response-source table and 97 were absent from all six. The latter define the strict response-source-disjoint subset. The cohort hashes, all exclusions and all teacher-source matches were committed before endpoint evaluation.

Both external endpoints were then fitted deployment-style on the same 1,280 external plus 85 ARROW labels with weights 1 and 3. They used the same descriptors, imputation, three 360-tree ensembles and seeds; only the 15 response columns differed. No Tier-A label entered fitting or selection. On N=220, structure-only and SolvAI MAEs were 1.53165 and 1.15255 kcal mol^{−1} (paired change -0.37909; 95% interval, -0.51692 to -0.24817). On strict N=97, the corresponding values were 2.13830 and 1.53560 kcal mol^{−1} (paired change -0.60271; interval, -0.86277 to -0.35858). The absolute errors delimit applicability and are not presented as PIMD8-level performance outside ARROW-85.

S11. Lambda-response probes

Short PIMD2 response fingerprints were available at $\lambda = 0.1, 0.5$ and 0.9 for 72, 74 and 73 molecules, respectively; 72 had complete three-point curves. The targets were the total solute–solvent $dH/d\lambda$ and its electrostatic and van der Waals components. A zero-valued polarization field was not interpreted. Structure-only heads were evaluated by OOF prediction, and their outputs were added to the same matched endpoint used for the corresponding campaign ablation. A separate direct-integration experiment used the predicted three-point curve with an affine calibration fitted inside each outer training fold. These targets were evaluated as candidate supervision and were not retained.

S12. Artifact verification

The public API accepts one or more SMILES strings. RDKit parses each input and emits canonical isomeric SMILES before feature generation. It does not standardize a query across tautomers, protonation states or salt forms. The runtime computes descriptors, evaluates the two D-MPNNs and four tree-artifact families, and passes the resulting 2,280-vector to three endpoint regressors. The artifact audit checks file hashes, feature count and names, software versions and two smoke predictions. It rejects features containing experimental target, PIMD, trajectory, probe, fold or scaffold information. The standardized-exclusion teacher refit is identified in the model card and manifest. The ensemble spread is returned for diagnosis but is not a calibrated per-query applicability or reliability score.

S13. Runtime benchmarking

Runtime was measured end to end from input SMILES to prediction on the released artifact. The host used an Intel Core i7-4930K CPU; the installed NVIDIA RTX 3090 was not used for inference. Cold single-molecule latency, five warm single-molecule runs, three batch-32 runs and peak resident memory were recorded. Chemprop predictor startup dominates single-molecule latency. These measurements describe the released implementation, not the irreducible model cost. The ARROW/PIMD8 comparison uses the published 15-window protocol and is kept separate from same-host inference timing.

Supplementary Notes

S1. Identity relationship to FreeSolv

Eighty of the 85 ARROW reference connectivities have a FreeSolv identity. This fact does not establish the provenance of every experimental value in the ARROW study. It instead determines the leakage rule: those connectivities were removed from all external supervised sources before endpoint or teacher fitting. The frozen SolvAI artifact is therefore not evaluated over unfiltered FreeSolv and no generic FreeSolv state-of-the-art claim is made.

S2. Interpretation of alternative supervision

Non-improving features are not interpreted as evidence that their underlying physical methods are inaccurate. A response can fail downstream because it is poorly predicted from structure, redundant with existing coordinates, mismatched to the endpoint or supported by too few labels. Compact ConfSolv summaries outperform larger graph and feed-forward latent representations in the matched exploratory screen. Sparse PIMD2 response produces physically recognizable trends but not the precision required for hydration-free-energy integration. These observations motivate the transfer condition stated in the main text: relevance, structural learnability and non-redundancy are all required.

Supplementary Tables

Supplementary Table 1 | SolvAI response priors

prior	name	physical meaning	source	units	surrogate	training rows	transformation	inference
1	combisolv_qm	COSMOtherm water response	CombiSolv-QM	kcal mol-1	CHEMELEON D-MPNN	3959	direct	structure surrogate; no simulation
2	abraham_e	excess molar refraction E	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
3	abraham_s	dipolarity/polarizability S	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
4	abraham_a	hydrogen-bond acidity A	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
5	abraham_b	hydrogen-bond basicity B	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
6	abraham_l	hexadecane-air partition L	SoluteML	Abraham scale	ExtraTrees	8098	direct	structure surrogate; no simulation
7	openff_corrected	explicit-water alchemical response	OpenFF 2.3.0 ASFE	kcal mol-1	ExtraTrees	520	prediction + residual	structure surrogate; no simulation
8	ghn2_corrected	implicit-solvent hydration response	GBa2	kcal mol-1	ExtraTrees	550	prediction + residual	structure surrogate; no simulation
9	smd_water	SMD water response	MolSolv	kcal mol-1	CHEMELEON D-MPNN	350359	direct	structure surrogate; no simulation
10	conf_gas_corr	gas conformer correction	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
11	conf_solution_corr	solution conformer correction	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
12	conf_hydration_corr	hydration conformer correction	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
13	conf_gsolv_sd	conformer solvation-energy spread	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
14	conf_response_mean	mean conformer solvent response	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation
15	conf_response_sd	conformer response spread	ConfSolv H2O	kcal mol-1	LightGBM	17829	direct	structure surrogate; no simulation

Supplementary Table 2 | Source-specific confirmatory filtering

source	unit	before standardized exclusion	removed	retained	role
CombiSolv-QM	unique structures	3961	2	3959	COSMOtherm water
MolSolv	SMD calculations	350391	32	350359	SMD(water)
ConfSolv	model-usable connectivities	17851	22	17829	conformer response
Endpoint labels	experimental connectivities	1280	0	1280	hydration free energy

Supplementary Table 3 | External endpoint-label sources

source group	selected records	selection rule
FreeSolv only	69	benchmark-disjoint connectivity
CombiSolv-Exp only	588	water records; benchmark-disjoint connectivity
FreeSolv + CombiSolv-Exp	490	one duplicate-resolved connectivity
SoluteML dGsolvDB3	133	at least two source measurements

Supplementary Table 4 | Complete repeated-partition values

Repeat	Split seed	Method	MAE
0	314159	A_structure_only	0.31780
0	314159	F_full_solvai	0.20788
1	271828	A_structure_only	0.31059
1	271828	F_full_solvai	0.21010
2	161803	A_structure_only	0.31309
2	161803	F_full_solvai	0.21217
3	141421	A_structure_only	0.31602
3	141421	F_full_solvai	0.20618
4	173205	A_structure_only	0.30727
4	173205	F_full_solvai	0.20053

Supplementary Table 5 | Global chemical-separation results

regime	method	n	mae
global_family	A_structure_only	85	1.262
global_family	F_full_solvai	85	0.468
global_scaffold	A_structure_only	85	0.728
global_scaffold	F_full_solvai	85	0.376
global_butina_0_70	A_structure_only	85	0.349
global_butina_0_70	F_full_solvai	85	0.216
global_nn_0.50	A_structure_only	85	0.359
global_nn_0.50	F_full_solvai	85	0.219
global_nn_0.60	A_structure_only	85	0.344
global_nn_0.60	F_full_solvai	85	0.211
global_nn_0.70	A_structure_only	85	0.343
global_nn_0.70	F_full_solvai	85	0.211
global_nn_0.80	A_structure_only	85	0.339
global_nn_0.80	F_full_solvai	85	0.210

Supplementary Table 6 | Diagnostic family errors

family	n	mae_kcal_mol
Acids	3	0.230
Alcohols	13	0.196
Aldehydes	3	0.117
Alkanes	13	0.234
Alkenes	5	0.085
Amides	4	0.483
Amines	5	0.138
Aromatics	13	0.313
Esters	4	0.055
Ethers	4	0.241
Heterorings	5	0.122
Ketones	5	0.160
Sulfides	4	0.115
Thiols	4	0.095

Supplementary Table 7 | Release-artifact hashes

file ₃₂₂	sha256	bytes
MODEL_CARD.md	a20bd031f5b85ba4f8fea701fa74f3d71f092ce79abb0c19b517711ff8ecfbde	1617
abraham_teacher.joblib	8c8a6365e7162dbe6ae445ce4004928ed86b2669d3e7a70d3d054f143ca97554	87055391
combisolv_qm.pt	2950be1232e2f868136e3388cf139e671d4196922ece0d3c7043d9dadece8b40	37452324
confsolv_teacher.joblib	a71487e53d5085e897dad02e62e95562933bc47262145fbf44286aae04b70b0d	5729365
head.joblib	fdc88cbac31696f254d85eb427b3c53a4811f29b3290d605b4f1ea66ebb95790	29424285
implicit_teacher.joblib	484d5e0c1efa759c8e7acbfcbdd87e218b93ae239e7f35303fe4cd7e310910f8	36316890
model_card.json	0f78177a701320a5aad77c45304ea73d5ca4799fb91e4f4fd01d0bb83f8e1244	1745
molsolv_smd.pt	35601030f030af72a9e6990bf8003df9d481858b27994b3fc70c81d605d130be	37226928
openff_teacher.joblib	55b7d31a293dd61fe57f5505e015775d7d86f53dca69980d398843706d58bb9b	48211661

Supplementary Table 8 | Runtime measurements

Measurement	Value
cold single molecule	15.312 s
warm single molecule median	15.294 s
warm single molecule p95	15.322 s
batch 32 median	15.824 s
batch 32 per molecule	0.494 s

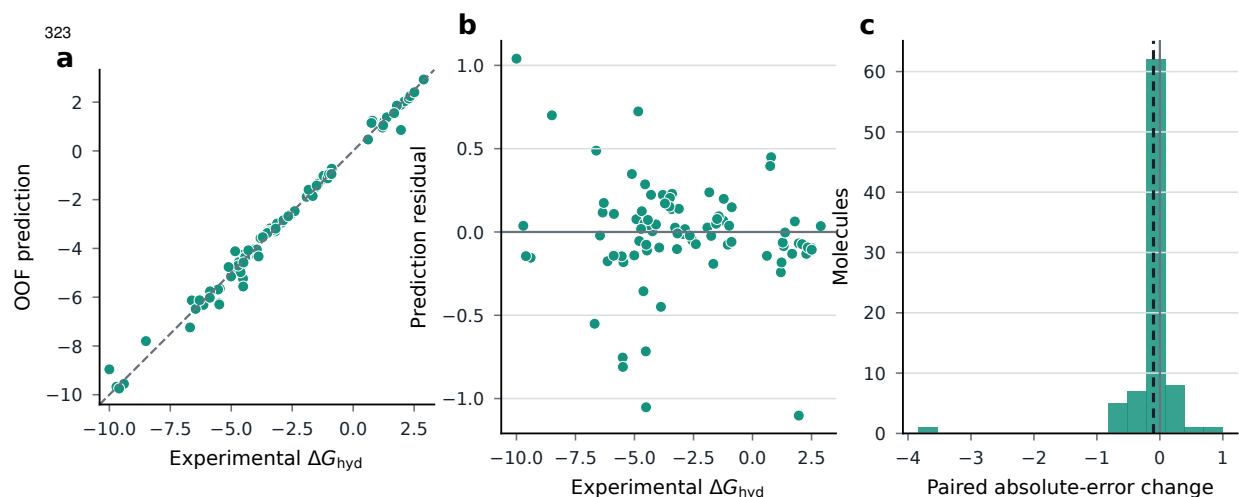
Supplementary Table 9 | Equal-weight ARROW sensitivity

Method	N	MAE	RMSE	Median absolute error
Matched structure-only	85	0.30977	0.59585	0.14972
Full SolvAI	85	0.20642	0.32627	0.11653

Supplementary Table 10 | Tier-A external validation

Cohort	Method	N	MAE	RMSE	Median absolute error
Endpoint-disjoint	Matched structure-only	220	1.53165	2.30974	0.96182
Endpoint-disjoint	Full SolvAI	220	1.15255	1.57723	0.84508
Strict response-source-disjoint	Matched structure-only	97	2.13830	3.08051	1.10237
Strict response-source-disjoint	Full SolvAI	97	1.53560	1.98792	1.24392

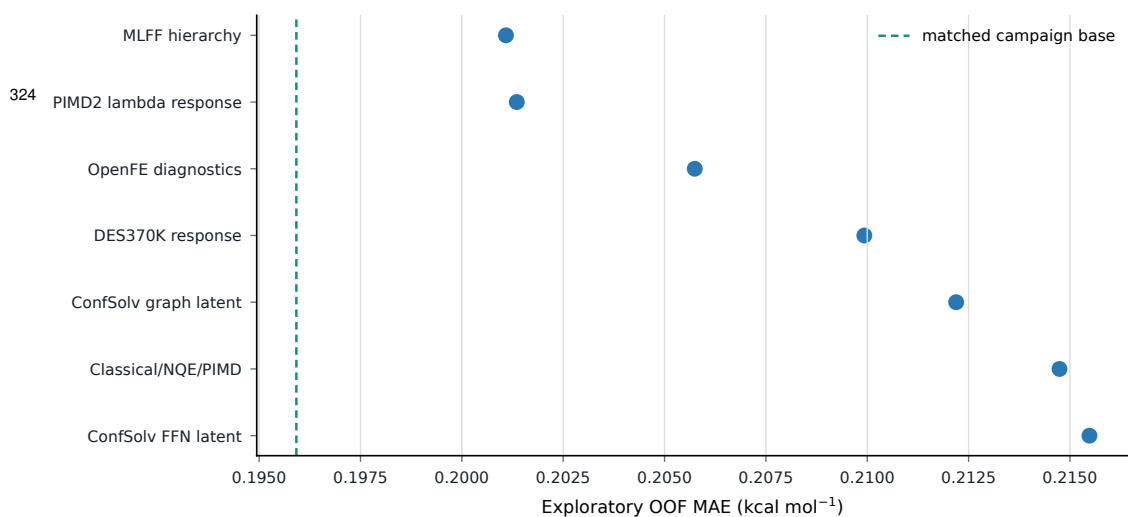
Supplementary Figures



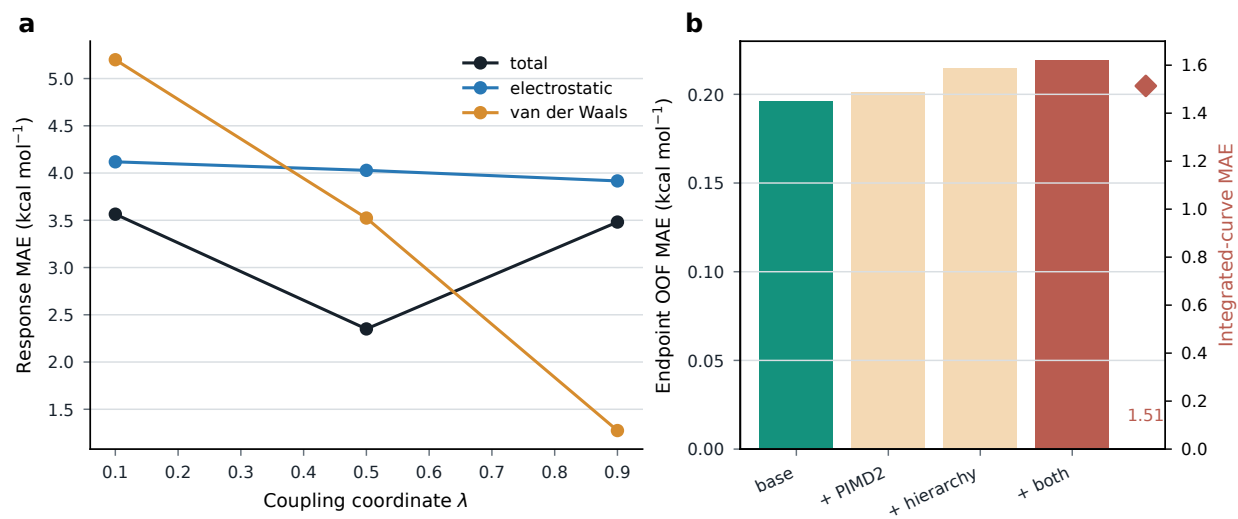
Supplementary Figure 1 | Molecule-level prediction and residuals. **a**, Confirmatory SolvAI OOF prediction against experiment. **b**, Signed residual against experimental hydration free energy. **c**, Distribution of the paired absolute-error change relative to the matched structure-only endpoint.



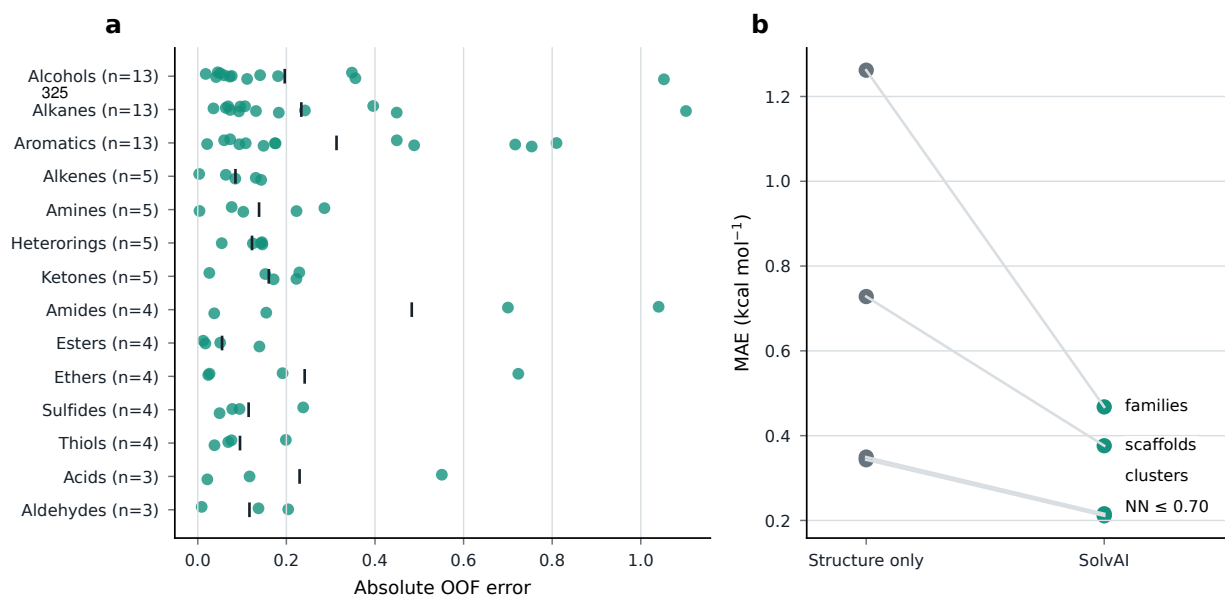
Supplementary Figure 2 | Source-specific provenance and standardized identity exclusion. Counts are shown in their native source units and normalized only within source. The audit removed standardized benchmark equivalents before the confirmatory teacher refits; no endpoint experimental label was removed.



Supplementary Figure 3 | Alternative physical-supervision routes. Exploratory fixed screens of additional diagnostic, latent and hierarchy features. The dashed line is the matched campaign response base for those screens. These experiments were not used to select the confirmatory controls.



Supplementary Figure 4 | Multi-λ response experiment. **a**, OOF response-head error at three coupling states. **b**, Matched endpoint consequences and the separately scaled error obtained by direct integration of the predicted curve.



Supplementary Figure 5 | Molecular and chemical-separation errors. **a**, Individual fixed-partition SolvAI errors grouped by diagnostic chemical family; ticks show family means and labels give sample sizes. **b**, Matched structure-only and SolvAI errors under four global separation regimes.

SIMBIOCHEM Submission Checklist

1. **Claims. Yes.** The abstract and introduction describe the matched response-layer contrast and distinguish ARROW-85 accuracy from external-cohort transfer.
2. **Limitations. Yes.** Section 5 states the molecular domain, external errors, missing applicability calibration, and the distinction from PIMD.
3. **AI / LLM disclosure. Yes.** Generative AI assisted with adapting and editing the manuscript for the workshop format. The authors reviewed the complete text and remain responsible for its correctness, originality, and integrity; AI assistance did not generate the scientific measurements reported here.
4. **Research integrity and originality. Yes.** This submission reports the distinct SolvAI response-prior study and is not a second version of another submission to this workshop.
5. **Reproducibility. Yes.** The appendix records data provenance, identity controls, models, folds, statistical procedures, and hashes; an anonymized artifact will provide frozen predictions and reproduction scripts.